College Event Statistics Portal

Mini Project — Data Analytics & Visualization

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Team size	Duration	Submission and presentation
2 students	**2–3 weeks**	**1st May 2026**

Background

Colleges run dozens of events every semester — technical fests, cultural programs, sports meets, workshops, and competitions — spread across departments. Currently, this data lives in scattered spreadsheets, notices, and emails, making it impossible to answer basic questions: Which department is most active? Which events see the highest student participation? What time of year are students most engaged?

Your task is to design and build a system that solves this — a centralized portal where event data is entered, stored, and turned into actionable insights for college stakeholders.

1. Problem Statement

Design and implement a College Event Intelligence Portal — a web-based system that allows event coordinators across departments to log event details, participants, and results in a structured manner, and allows stakeholders to explore analytics and visualizations drawn from that data.

2. System Requirements

2.1 Data Entry Portal

Build a web interface through which authorized event coordinators can submit the following:

- Event details — name, department, category (technical / cultural / sports / workshop / other), date, venue, description, organizer name
- Participant details — participant name, roll number, department, year of study, whether internal or external participant
- Competition results — if the event is a competition, store rank/position, award/prize if any
- Flexible event types — the schema must gracefully handle events that have no competition (e.g. workshops) and events that do (e.g. hackathons). Design for this variability.

2.2 Database Design

Model and implement a relational database that:

- Has clearly defined tables with primary keys and foreign keys
- Avoids redundancy — normalized to at least 2NF
- Supports different event types without breaking schema integrity
- Logs submission timestamp for every record
- Maintains separation between raw event data and derived/aggregated data
- Supports efficient time-based and department-based querying

2.3 Data Processing Layer

Implement a processing pipeline (Python) that:

- Cleans incoming data — handles nulls, duplicates, inconsistent formatting
- Standardizes categorical fields (e.g. department names, event categories)
- Computes derived metrics such as: events per department, participation rate, top performers, event frequency by month
- Is repeatable — running it multiple times must produce the same result

2.4 Analytics & Visualization

Provide insights for the following stakeholder views. These can be delivered as a dashboard within the website, an embedded Power BI report, or a separate visualization page — your choice.

(Given insights are only for the reference, students may customize them and justify them if necessary)

Stakeholder	Required insights
HODs & Management	Department-wise event count, participation volume, inter-department comparison, semester trends
Faculty / Student Affairs	Event calendar view, category breakdown, internal vs external participation ratio, active organizers
Students	Upcoming and past events, results lookup, individual participation history by roll number
All stakeholders	Filter by date range, department, event category. At least one derived/composite metric beyond raw counts.

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3. Technical Constraints

No tech stack is mandated, but the following course concepts must be demonstrably used in the submission:

- Relational database design & SQL
- Python data processing

- Data cleaning & normalization
- Visualization (any tool — Power BI, Seaborn, Matplotlib, or embedded in the website)
- Schema design with ER diagram
- Analytical SQL queries

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4. Deliverables

- Working web portal (hosted locally or deployed)
- Database schema diagram (ER diagram) with justification
- SQL scripts — schema creation + at least 5 meaningful analytical queries
- Python processing scripts with comments
- Dashboard / visualization (embedded or standalone)
- Final report — architecture, design decisions, sample data used, insights found

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5. Hosting & Deployment Constraints

The application must be fully deployed and accessible via a public URL by the time of submission. A demo that runs only on localhost will not be accepted.

5.1 Application Hosting

- The web application must be hosted on a cloud platform and accessible via a public URL
- Free-tier platforms are perfectly acceptable — recommended options include Render, Vercel, Railway, Netlify, or any equivalent
- The public URL must remain live and accessible through the evaluation period
- The submitted report must include the public URL and instructions to access the portal

5.2 Database Hosting

- The database must be cloud-hosted — a locally running database is not acceptable
- Recommended free-tier options: Supabase (PostgreSQL), PlanetScale (MySQL), Railway (PostgreSQL/MySQL), Neon (PostgreSQL), or MongoDB Atlas (NoSQL)
- The application must connect to the cloud database at runtime, not a locally exported file
- Database credentials must be managed via environment variables and must not be hardcoded in source code

5.3 Source Code

- Source code must be submitted via a public GitHub repository link
- The repository must include a README with setup instructions, the public URL, and the architecture diagram

6. Evaluation Rubric

Component	Marks
Database schema design & normalization	20
Data entry portal — completeness & flexibility	20
Data processing & cleaning pipeline	15
Analytical SQL queries	15
Visualization & dashboard quality	20
Report & presentation	10
Total	**100**

NOTE

Students can use the data of recent college events like Samyuthi and Ojas — including event details, participants, and results — to populate their portal and demonstrate the effectiveness of their project. Using real, familiar data will make your visualizations more meaningful and your demo more convincing.

Be cautious about confidentiality of the data and take proper guidance from faculty coordinators whenever needed.